ME6030 - Markov Matrix Analysis

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1 Problem Setup

A manufacturing plant operates three workstations:

State	Workstation	Description
1	Machining Centre	Primary processing of incoming jobs
2	Quality Inspection Bay	Inspection after machining
3	Rework Station	Correction of defective parts

The system transitions between states at the end of each production cycle according to the **Markov transition matrix** P, where P_{ij} denotes the probability of transitioning from State j to State i in one cycle:

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0 & 0.3 & 0.7 \end{bmatrix} \tag{1}$$

Every column sums to 1 and all entries are non-negative, so P is a valid column-stochastic Markov matrix. The governing **difference equation** is:

$$\mathbf{z}_{k+1} = P\,\mathbf{z}_k, \qquad \text{with solution} \quad \mathbf{z}_k = P^k\,\mathbf{z}_0 \tag{2}$$

where $\mathbf{z}_0$ is the initial state distribution.

2 Proof: $\lambda = 1$ is Always an Eigenvalue of a Markov Matrix

Theorem 1. *Let $P \in \mathbb{R}^{n\times n}$ be column-stochastic, i.e. $P_{ij} \geq 0$ and $\sum_{i=1}^{n} P_{ij} = 1$ for all j. Then $\lambda = 1$ is an eigenvalue of P.*

Proof. Let $\mathbf{1} = [1,\, 1,\, \ldots,\, 1]^T \in \mathbb{R}^n$. The column-stochastic condition says every column of P sums to 1, which is exactly:

$$\mathbf{1}^T P = \mathbf{1}^T \tag{3}$$

because the j-th entry of $\mathbf{1}^T P$ is $\sum_{i=1}^{n} P_{ij} = 1$. Equation (3) says $\mathbf{1}^T$ is a left eigenvector of P with eigenvalue 1. Transposing both sides:

$$P^T\mathbf{1} = \mathbf{1} \tag{4}$$

so $\mathbf{1}$ is a right eigenvector of P^T with eigenvalue 1. Since P and P^T share the same characteristic polynomial,

$$\det(P - \lambda I) = \det\big((P - \lambda I)^T\big) = \det(P^T - \lambda I), \tag{5}$$

they have identical eigenvalues. Therefore $\lambda = 1$ is also an eigenvalue of P. A corresponding right eigenvector $\boldsymbol{\pi}$ satisfying $P\boldsymbol{\pi} = \boldsymbol{\pi}$ exists; by the Perron–Frobenius theorem it has strictly positive entries for an irreducible positive stochastic matrix. □

3 Eigenvalues of P

Characteristic Polynomial

The eigenvalues satisfy $\det(P - \lambda I) = 0$. Expanding the determinant:

$$\det \begin{bmatrix} 0.7-\lambda & 0.2 & 0.1 \\ 0.3 & 0.5-\lambda & 0.2 \\ 0 & 0.3 & 0.7-\lambda \end{bmatrix} = 0 \tag{6}$$

Expanding along the first column and simplifying:

$$-\lambda^3 + 1.9\lambda^2 - 1.07\lambda + 0.17 = 0 \quad\Longleftrightarrow\quad \lambda^3 - 1.9\lambda^2 + 1.07\lambda - 0.17 = 0 \tag{7}$$

Since $\lambda = 1$ is a guaranteed root (Theorem 1), we factor out $(\lambda - 1)$:

$$(\lambda - 1)\big(\lambda^2 - 0.9\lambda + 0.17\big) = 0 \tag{8}$$

The quadratic $\lambda^2 - 0.9\lambda + 0.17 = 0$ has discriminant $\Delta = 0.9^2 - 4(0.17) = 0.81 - 0.68 = 0.13 > 0$, giving two real roots:

$$\lambda_{2,3} = \frac{0.9 \pm \sqrt{0.13}}{2} \tag{9}$$

Numerical Values

	Eigenvalue	$\lvert\lambda_i\rvert$	**Note**
λ_1	1.000000	1.000000	Steady-state mode (guaranteed)
λ_2	$0.630278 = \dfrac{0.9 + \sqrt{0.13}}{2}$	0.630278	Transient mode
λ_3	$0.269722 = \dfrac{0.9 - \sqrt{0.13}}{2}$	0.269722	Transient mode

All eigenvalues are real. Since $|\lambda_2|, |\lambda_3| < 1$, the transient modes decay geometrically and the system converges to a unique steady state.

4 Eigenvectors of P

Eigenvector for $\lambda_1 = 1$ (Analytical)

We solve $(P - I)\mathbf{v} = \mathbf{0}$:

$$P - I = \begin{bmatrix} -0.3 & 0.2 & 0.1 \\ 0.3 & -0.5 & 0.2 \\ 0 & 0.3 & -0.3 \end{bmatrix} \tag{10}$$

Row 3 gives $0.3\,v_2 - 0.3\,v_3 = 0$, so $v_2 = v_3$. Row 1 then gives $-0.3\,v_1 + 0.2\,v_2 + 0.1\,v_2 = 0$, i.e. $-0.3\,v_1 + 0.3\,v_2 = 0$, so $v_1 = v_2$. Hence $v_1 = v_2 = v_3$ and the eigenvector is proportional to $\mathbf{1}$:

$$\mathbf{v}_1 \propto \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \tag{11}$$

Eigenvectors for λ_2 and λ_3 (Numerical)

The remaining eigenvectors are obtained via `numpy.linalg.eig` (L2-normalised):

$$\mathbf{v}_2 \approx \begin{bmatrix} 0.5988 \\ 0.1813 \\ -0.7801 \end{bmatrix}, \qquad \mathbf{v}_3 \approx \begin{bmatrix} 0.2410 \\ -0.7961 \\ 0.5551 \end{bmatrix} \tag{12}$$

Both $\mathbf{v}_2$ and $\mathbf{v}_3$ have mixed signs, meaning they represent oscillatory redistribution between workstations that decays over time.

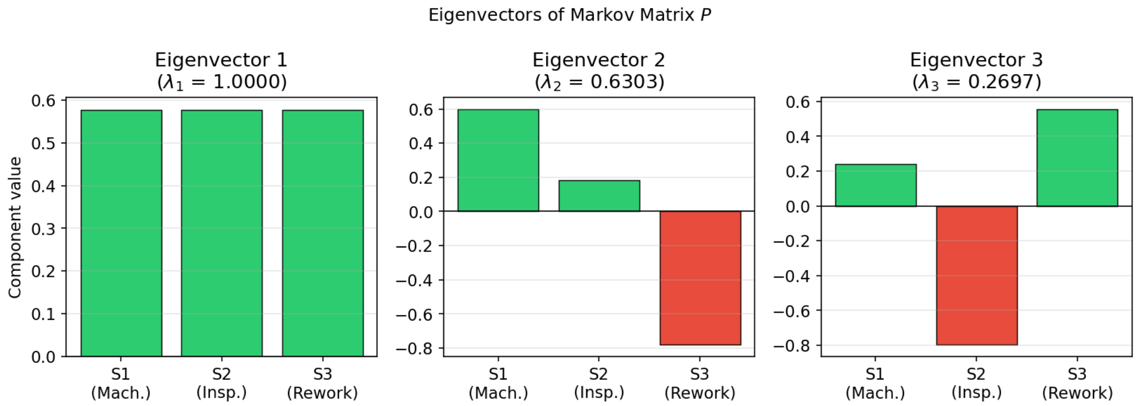

Figure 1: Components of all three eigenvectors. Eigenvector 1 (all equal positive bars) is the uniform steady-state mode. Eigenvectors 2 and 3 have mixed signs and represent transient redistribution that decays geometrically.

5 The $\lambda = 1$ Eigenvector and Its Physical Significance

L1-normalising $\mathbf{v}_1 \propto [1, 1, 1]^T$ so that its components sum to 1 gives the **steady-state distribution**:

$$\boldsymbol{\pi} = \frac{1}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \approx \begin{bmatrix} 0.3333 \\ 0.3333 \\ 0.3333 \end{bmatrix} \tag{13}$$

This is an exact result confirmed analytically: $v_1 = v_2 = v_3$ forces a perfectly uniform distribution. One can verify $P\boldsymbol{\pi} = \boldsymbol{\pi}$ directly since each row of P sums to $0.7+0.2+0.1 = 1$, $0.3+0.5+0.2 = 1$, $0+0.3+0.7 = 1$, so multiplying P by $[1/3,\ 1/3,\ 1/3]^T$ returns the same vector.

Physical interpretation: In the long run, the manufacturing system spends **exactly equal time at every workstation** one-third of all production cycles at the Machining Centre, one-third at the Inspection Bay, and one-third at the Rework Station, regardless of initial conditions. This uniform distribution arises because the transition probabilities, while asymmetric, are balanced in a way that produces no preferred long-run state.

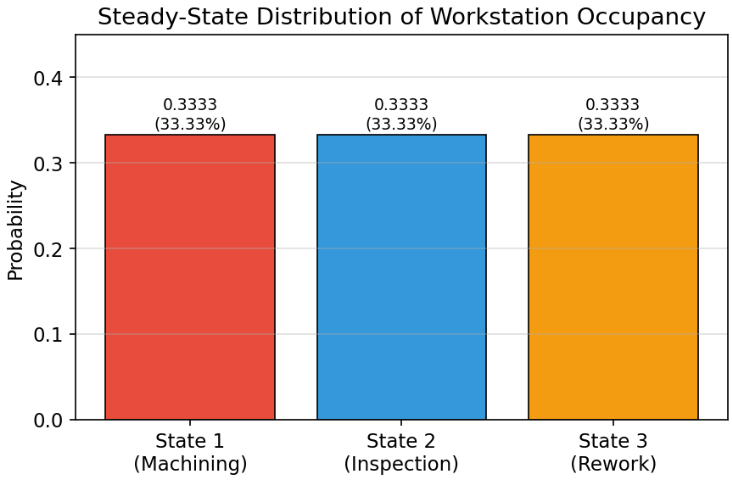

Figure 2: Steady-state workstation occupancy $\boldsymbol{\pi} = [1/3,\ 1/3,\ 1/3]^T$. Each workstation is occupied exactly one-third of the time in the long run.

6 Eigenvalue Magnitudes and Convergence Rate

$\lvert\lambda_1\rvert$	$\lvert\lambda_2\rvert$	$\lvert\lambda_3\rvert$
1.0000	0.6303	0.2697

Key observations:

- $|\lambda_1| = 1$: the steady-state mode persists and defines the long-run behaviour.
- $|\lambda_2| = 0.6303 < 1$: this transient mode decays as 0.6303^k. After 10 steps it retains only $0.6303^{10} \approx 0.7\%$ of its initial amplitude; after 20 steps less than 0.005%.
- $|\lambda_3| = 0.2697 < 1$: even faster decay; after 5 steps it retains only $0.2697^5 \approx 0.14\%$.
- The **spectral gap** $1 - |\lambda_2| = 0.3697$ governs the worst-case rate of convergence. A larger gap means faster convergence.

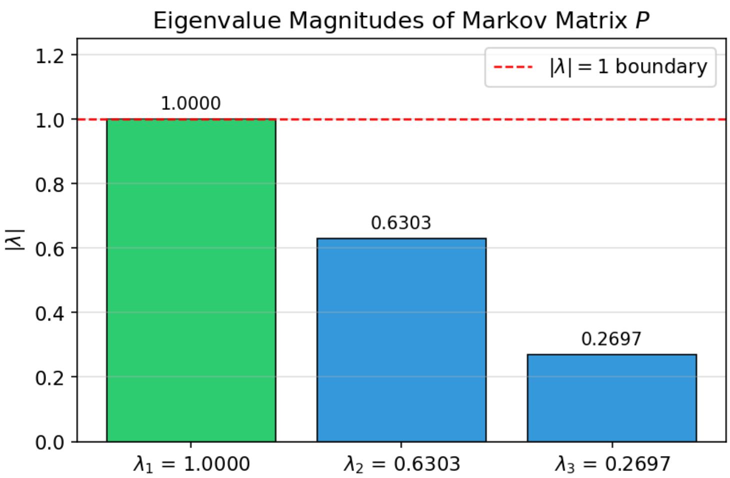

Figure 3: Magnitudes of the three eigenvalues. $\lambda_1 = 1$ (green) lies on the unit-circle boundary; λ_2 and λ_3 (blue) are strictly inside, ensuring asymptotic convergence to the steady state.

7 Steady-State Analysis: $\mathbf{z}_k$ for Large k

Analytical Result

Any initial distribution $\mathbf{z}_0$ decomposes into the eigenbasis:

$$\mathbf{z}_0 = c_1\,\mathbf{v}_1 + c_2\,\mathbf{v}_2 + c_3\,\mathbf{v}_3 \tag{14}$$

Applying P repeatedly and using $P\mathbf{v}_i = \lambda_i \mathbf{v}_i$:

$$\mathbf{z}_k = P^k \mathbf{z}_0 = c_1 \underbrace{\lambda_1^k}_{=1} \mathbf{v}_1 + c_2 \underbrace{(0.6303)^k}_{\to 0} \mathbf{v}_2 + c_3 \underbrace{(0.2697)^k}_{\to 0} \mathbf{v}_3 \tag{15}$$

As $k \to \infty$ both transient terms vanish:

$$\lim_{k\to\infty} \mathbf{z}_k = c_1\,\mathbf{v}_1 = \boldsymbol{\pi} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \tag{16}$$

The matrix power converges to the rank-1 matrix $P^\infty = \boldsymbol{\pi}\,\mathbf{1}^T$, i.e. every column of P^k tends to $\boldsymbol{\pi}$. The long-run state is therefore **independent of the initial condition** $\mathbf{z}_0$.

Numerical Verification ($k = 10\,000$)

Starting from $\mathbf{z}_0 = [1,\,0,\,0]^T$ (all jobs initially at the machining centre) and iterating $\mathbf{z}_{k+1} = P\,\mathbf{z}_k$ for 10 000 steps:

$$\mathbf{z}_{10000} = \begin{bmatrix} 0.33333333 \\ 0.33333333 \\ 0.33333333 \end{bmatrix}, \qquad |\mathbf{z}_{10000} - \boldsymbol{\pi}| < 10^{-15} \tag{17}$$

The simulation confirms convergence to machine-precision accuracy, consistent with the analytical prediction.

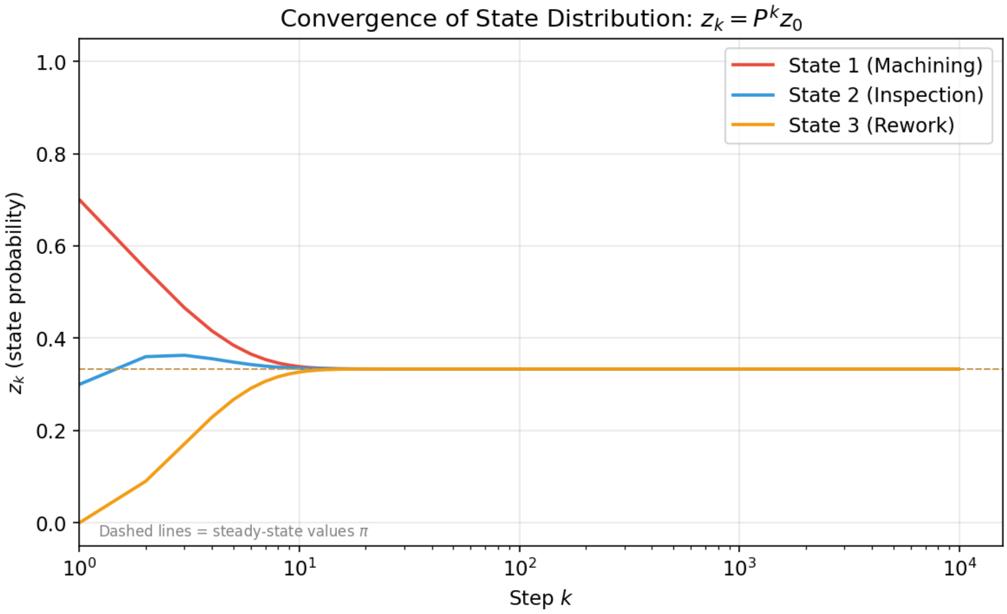

Figure 4: Evolution of $\mathbf{z}_k$ over 10 000 steps (log scale on k). All three state probabilities converge to 1/3 (dashed lines). Convergence is essentially complete by $k \approx$ 15–20 steps, consistent with the spectral gap of $1 - |\lambda_2| = 0.3697$.